

## 51BM0.25/SIP-EJ444-E3-X2

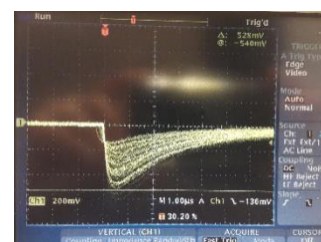
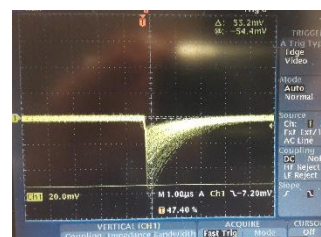
**Description:** Thin model **Alpha/Beta** SiPM Scintillation detector with built in temperature compensated bias generator and preamplifier

|                               |                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Scintillation material</b> | : ZnS(Ag)+plastic scintillator<br>Ø51 x 0.25mm (EJ-444)                                              |
| <b>Drawing number</b>         | : V-1611                                                                                             |
| <b>Crystal read out</b>       | : SiPm, Array J-60035-4P                                                                             |
| <b>Housing</b>                | : black light tight polyester                                                                        |
| <b>Entrance window</b>        | : light-tight double aluminized<br>mylar 0.9 mg/cm <sup>2</sup>                                      |
| <b>Protection</b>             | : 72 % transmission stainless steel grid                                                             |
| <b>Power supply</b>           | : + 5 – 16 V (3 mA)                                                                                  |
| <b>Electrical connections</b> | : LEMO connector ERA0S302CLL Flying leads<br><br>Pin 1 = +5 – 16 V (brown)<br>Pin 2 = Signal (white) |
| <b>Cable</b>                  | : 50 cm 4 core shielded LIYD11Y                                                                      |

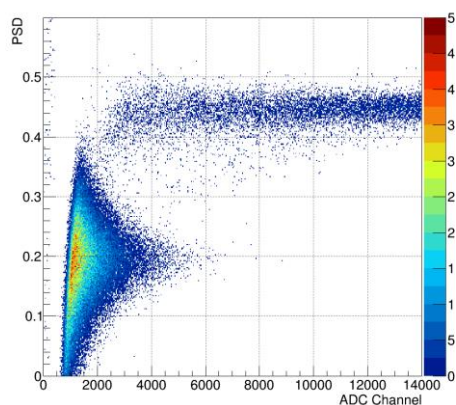
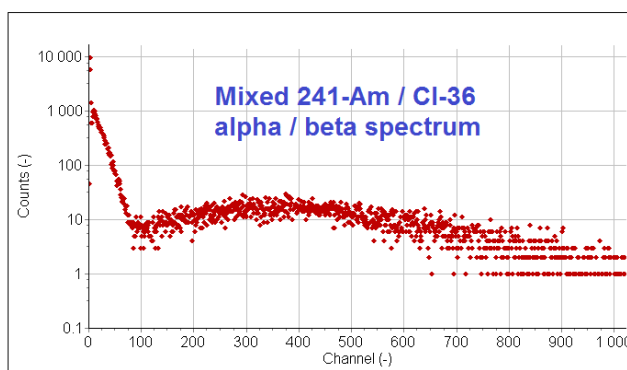
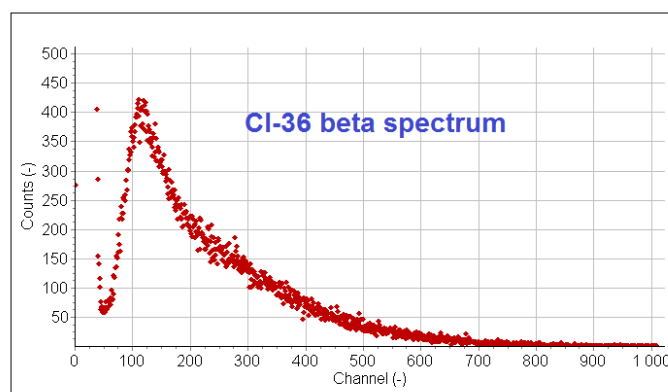
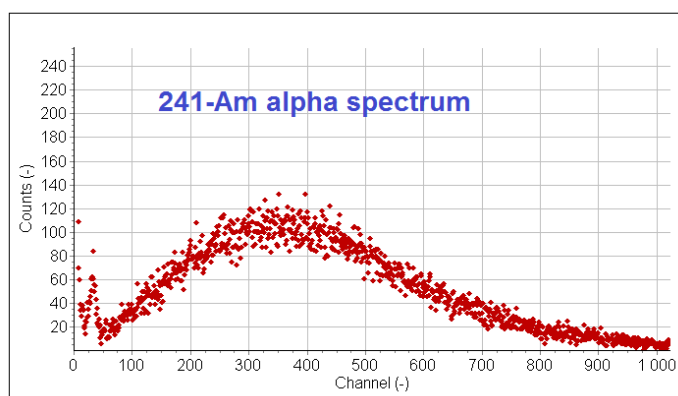
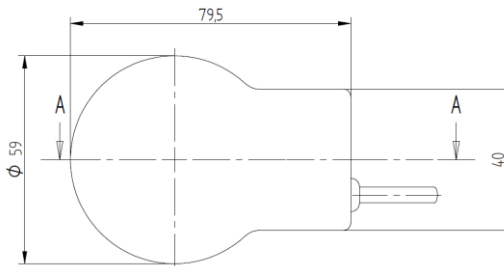
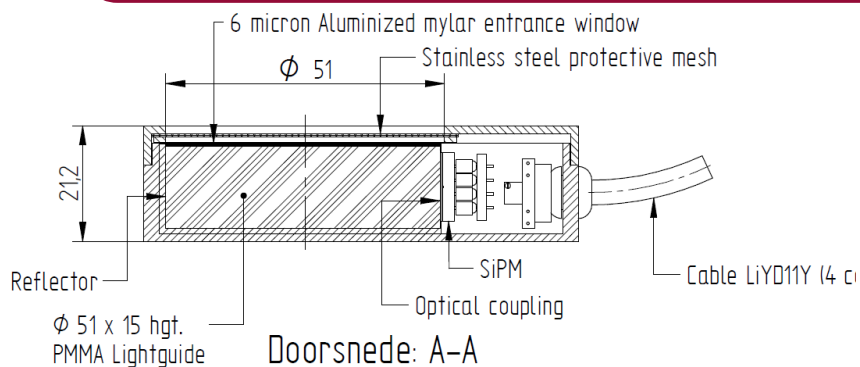


### Performance

|                                 |                                                                |
|---------------------------------|----------------------------------------------------------------|
| <b>Pulse rise time (0-100%)</b> | : 150 ns (betas)<br>530 ns (alphas)                            |
| <b>Pulse fall time (1/e)</b>    | : 1.2 µs (betas)<br>530 ns (alphas)                            |
| <b>Gain</b>                     | : Approx. 75 mV/MeV(betas)<br>Approx. 500mV /5.48 MeV (alphas) |
| <b>Noise level</b>              | : < 50 keV (betas) @ 22 degrees C                              |



Last revision: 08-02-2019  
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2D PSD scatterplots of detector under mixed alpha/beta irradiation